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RESEARCH ARTICLE

Relationship between peroxisome proliferator-activated receptor alpha activity and cellular concentration of 14 perfluoroalkyl substances in HepG2 cells

Anna Kjerstine Rosenmai[1] | Lutz Ahrens[2] | Théo le Godec[1] | Johan Lundqvist[1] | Agneta Oskarsson[1]

[1] Department of Biomedical Sciences and Veterinary Public Health, Swedish University of Agricultural Sciences, Box 7028, SE-750 07 Uppsala, Sweden

[2] Department of Aquatic Sciences and Assessment, Swedish University of Agricultural Sciences, Box 7050, SE-750 07 Uppsala, Sweden

Correspondence
Anna Kjerstine Rosenmai, Department of Biomedical Sciences and Veterinary Public Health, Swedish University of Agricultural Sciences, Box 7028, SE-750 07 Uppsala, Sweden.
Email: anna.rosenmai@slu.se

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Abstract

Peroxisome proliferator-activated receptor alpha (PPARα) is a molecular target for perfluoroalkyl substances (PFASs). Little is known about the cellular uptake of PFASs and how it affects the PPARα activity. We investigated the relationship between PPARα activity and cellular concentration in HepG2 cells of 14 PFASs, including perfluoroalkyl carboxylates (PFCAs), perfluoroalkyl sulfonates and perfluorooctane sulfonamide (FOSA). Cellular concentrations were determined by high-performance liquid chromatography–tandem mass spectrometry and PPARα activity was determined in transiently transfected cells by reporter gene assay. Cellular uptake of the PFASs was low (0.04–4.1%) with absolute cellular concentrations in the range 4–2500 ng mg^{-1} protein. Cellular concentration of PFCAs increased with perfluorocarbon chain length up to perfluorododecanoate. PPARα activity of PFCAs increased with chain length up to perfluorooctanoate. The maximum induction of PPARα activity was similar for short-chain (perfluorobutanoate and perfluoropentanoate) and long-chain PFCAs (perfluorododecanoate and perfluorotetradecanoate) (approximately twofold). However, PPARα activities were induced at lower cellular concentrations for the short-chain homologs compared to the long-chain homologs. Perfluorohexanoate, perfluoroheptanoate, perfluorooctanoate, perfluorononanoate (PFNA) and perfluorodecanoate induced PPARα activities >2.5-fold compared to controls. The concentration–response relationships were positive for all the tested compounds, except perfluorooctane sulfonate PFOS and FOSA, and were compound-specific, as demonstrated by differences in the estimated slopes. The relationships were steeper for PFCAs with chain lengths up to and including PFNA than for the other studied PFASs. To our knowledge, this is the first report establishing relationships between PPARα activity and cellular concentration of a broad range of PFASs.

KEYWORDS

cellular concentration, perfluoroalkane sulfonamide, perfluoroalkyl carboxylates, perfluoroalkyl sulfonates, peroxisome proliferator-activated receptor alpha, PPARα

1 | INTRODUCTION

Perfluoroalkyl substances (PFASs) have been used in a variety of consumer products, such as surface treatment of textiles, leather, carpets and food contact materials (Wang, Cousins, Scheringer, & Hungerbühler, 2013). PFASs are aliphatic substances in which all carbons are fully fluorinated, except the carbon in the functional group (Buck et al., 2011) (Figure 1). Examples are the well-known perfluorooctanoate (PFOA) and perfluorooctane sulfonate (PFOS) belonging to the perfluoroalkyl carboxylates (PFCAs) and perfluoroalkyl sulfonates (PFSAs), respectively, but homologs of these exist with shorter or longer perfluorocarbon chains.

PFASs are ubiquitously spread in the environment (Ahrens, 2011; Giesy & Kannan, 2002; Houde, De Silva, Muir, & Letcher, 2011; McLachlan, Holmström, Reth, & Berger, 2007; Zareitalabad, Siemens, Hamer, & Amelung, 2013) and are detected in human blood worldwide (CDC, 2015; Fromme, Tittlemier, Völkel, Wilhelm, & Twardella, 2009). In the period 1999–2008, PFOA, PFOS, perfluorohexane sulfonate

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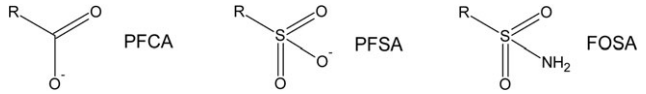


FIGURE 1 Chemical structures of perfluoroalkyl carboxylate (PFCA), perfluoroalkyl sulfonate (PFSA) and perfluorooctane sulfonamide (FOSA). R = $F(CF_2)_n$, represent the perfluorocarbon chain of the tested PFASs with n = 3–11 and 13 for the PFCAs, n = 4, 6 and 8 for the PFSAs and n = 8 for the FOSA

(PFHxS) and perfluorononanoate (PFNA) have consistently been detected in more than 94% of serum samples in the US population (n = 1562–2120) (Kato, Wong, Jia, Kuklenyik, & Calafat, 2011) and geometric mean concentrations of 2.08, 6.31, 1.28 and 0.881 μg l^{-1}, respectively, have been reported in serum samples from 2011 to 2012 (CDC, 2015). Some PFASs have long serum half-lives, e.g. ~8.5 years for PFHxS (Olsen et al., 2007) and thus once exposure has occurred a long residence time is expected in humans.

PFASs have been reported to have adverse effects in animal studies and particularly the toxicity of PFOA and PFOS is well documented (Andersen et al., 2008; Kennedy et al., 2004; Lau et al., 2007; Lau, Butenhoff, & Rogers, 2004). Among the reported effects are hepatotoxicity, including hepatomegaly through hepatocytic hypertrophy, induced peroxisome β-oxidation and tumor development (Lau, 2012). Though the mechanisms behind the toxicity is not well understood, one molecular target for PFASs is the peroxisome proliferator-activated receptor alpha (PPARα), which is supported by in vitro data (Buhrke, Kibellus, & Lampen, 2013; Rosenmai et al., 2016; Takacs & Abbott, 2007; Wolf, Schmid, Lau, & Abbott, 2012; Wolf, Takacs, Schmid, Lau, & Abbott, 2008). PPARα activities are dependent on the cellular concentration and affinity to the receptor of PFASs. Surprisingly little information is available on the cellular uptake of PFASs, the impact of chain length and functional groups on cellular uptake, and the relationship between biological activity and cellular uptake.

Regulatory measures have been taken to ban or reduce some PFASs, which has led to a search for substitutes such as short-chain PFASs (Scheringer et al., 2014; Wang et al., 2013). However, the short-chain homologs are less investigated, and thus further information on their potential toxicity is needed.

The aim of this study was to investigate the relationship between PPARα activity and cellular concentration of PFASs. We tested 14 PFASs of varying perfluorocarbon chain lengths and functional groups, including 10 PFCAs, three PFSAs, as well as perfluorooctane sulfonamide (FOSA) (Figure 1). All the tested PFASs were detected in the cells, most but not all PFASs induced PPARα activity, there was generally a statistically significant relationship between the activity and the cellular concentration, and the cellular concentration as well as PPARα activity was dependent on chain length and functional group.

2 | MATERIALS AND METHODS

2.1 | Test chemicals

In total, 14 PFASs were included for analysis of in vitro activity and cellular concentration; perfluorobutanoate (PFBA, 98%), perfluoropentanoate (PFPeA, 97%), perfluorohexanoate (PFHxA, ≥97%), perfluoroheptanoate (PFHpA, 99%), PFOA (≥98%), PFNA (97%), perfluorodecanoate (PFDA, 98%), perfluoroundecanoate (PFUnDA, 95%), perfluorododecanoate (PFDoDA, 95%), perfluorotetradecanoate (PFTeDA, 97%), perfluorobutane sulfonate (PFBS, purity 98%), PFHxS (≥98%), PFOS (98%) and FOSA (99%) (purchased from Sigma-Aldrich, St. Louis, MO, USA) (Supporting information, Table S1). PFASs used as standards for chemical analysis were purchased from Wellington Laboratories (Guelph, ON, Canada). An internal standard mixture was used for the determination of cellular concentrations containing mass-labeled $^{13}C_8$-FOSA, $^{13}C_4$-PFBA, $^{13}C_2$-PFHxA, $^{13}C_4$-PFOA, $^{13}C_5$-PFNA, $^{13}C_2$-PFDA, $^{13}C_2$-PFUnDA, $^{13}C_2$-PFDoDA, $^{18}O_2$-PFHxS and $^{13}C_4$-PFOS (purchased from Wellington Laboratories). PFAS stock solutions for *in vitro* investigations were prepared in ethanol in concentrations ranging from 4 to 350 mM depending on compound solubility.

2.2 | Cell culture

Human hepatocellular carcinoma (HepG2) cells were cultured in Dulbecco's modified Eagle medium (Lonza, Basel, Switzerland) supplemented with 2 mM L-glutamine (Gibco, Thermo Fisher Scientific, Waltham, MA, USA), an antibiotic–antimycotic solution with a final concentration of 100 U ml^{-1} penicillin, 100 μg ml^{-1} streptomycin, and 0.25 μg ml^{-1} Gibco amphotericin B (Gibco, Thermo Fisher Scientific) and 10% fetal bovine serum (FBS) (Gibco, Thermo Fisher Scientific). The cell culture was kept in a humidified atmosphere at 37°C containing 95% air and 5% CO_2.

2.3 | Cell viability

Cell viability was tested by the MTS-based colorimetric assay (Cell Titer 96® Aqueous One Solution Cell Proliferation Assay; Promega, Madison, WI, USA). The formation of the colored product formazan from a tetrazolium compound (MTS) is measured, which is directly proportional to the number of viable cells in culture. HepG2 cells were seeded in transparent 96-well plates, 1.5 × 10^4 cells per well, medium was changed after 48 hours and treatment with 0.1–100 μM PFASs was initiated 24 hours later. After 24 hours treatment, 20 μl per well of cell viability assay reagent was added, cells were incubated for approximately 15 minutes and absorbance was measured on a Wallac Victor 1420 microplate reader (PerkinElmer, Waltham, MA, USA) at 490 nm. A decrease in viability of ≥20% was considered as cytotoxic and such concentrations of test compounds were not included in the following experiments.

2.4 | Peroxisome proliferator-activated receptor alpha activity

The PPARα activity was determined by a reporter gene assay in HepG2 cells. Cells were seeded in 48-well plates (Sarstedt, North Rhine-Westphalia, Nümbrecht, Germany) with 3.75 × 10^4 cells per well and cultured for 48 hours. After changing the medium, cells were transiently transfected using 0.6 μl per well of Lipofectamine (Invitrogen, Thermo Fisher Scientific, Waltham, MA, USA) in accordance with the manufacturer's protocol. The plasmids used were a human PPARα ligand binding domain-GAL4-plasmid (75 ng per well), a Gal4-

responsive luciferase expressing reporter plasmid (UAS-Gal-Luc) (75 ng per well) (both donated by Dr. Jan Fleckner, Novo Nordic, Denmark), and a renilla luciferase plasmid (50 ng per well) as a control for the transfection efficiency. Following transfection, cells were incubated for 24 hours before treatment with individual PFASs in concentrations of 10, 30 and 100 μM for 24 hours. Ethanol was used as control and the ethanol concentration was constant across all treatments and controls at 3% ethanol, a concentration that did not affect cell viability. In addition, a known PPARα agonist, WY-14643, was included as a positive control. The experiments were finalized with medium removal and cell lysis with 65 μl of passive lysis buffer (Promega) for 15–30 minutes. Two aliquots of 2.5 μl cell lysate were sampled for protein quantification and 20 μl was transferred to white 96-well plates (Corning, NY, USA). Luciferase activity was assayed using the Dual-Luciferase® Reporter Assay System (Promega) according to the manufacturer's protocol. The substrate for the firefly luciferase was added, 45 μl, and the luminescence was measured on a Wallac Victor 1420 microplate reader. Successively, 45 μl substrate was added to allow for renilla luciferase activity measurement.

2.5 | Cellular perfluoroalkyl substance quantification

For determination of cellular PFASs concentration HepG2 cells were seeded in 12-well plates (Corning) at a concentration of 18×10^4 cells per well. Medium was changed after 48 hours and after another 24 hours, cells were exposed to PFASs in concentrations of 10, 30 and 100 μM. After 24 hours of treatment with PFASs, cells were washed 5 times with ice-cold phosphate-buffered saline after which cells were lysed with 250 μl passive lysis buffer (from the Dual-Luciferase® Reporter Assay System; Promega) and lysates from three wells per treatment were pooled. An aliquot of the pooled sample was used for protein quantification and the lysate for PFAS quantification was centrifuged at 16 000 *g* for 15 minutes. The supernatant was collected in Eppendorf tubes and stored at −80°C until PFAS analysis.

Fifty μl of the cell lysate was transferred into a 1.5 ml brown glass vial, and 100 μl internal standard mixture (c = 50 ng ml^{-1}) as well as 850 μl methanol were added. Vials were subsequently vortexed for 30 seconds. The separation and detection of PFASs were performed by high-performance liquid chromatography (Agilent Technologies 1200 Series, Palo Alto, CA, USA) coupled to a triple quadrupole mass spectrometer interfaced with an electrospray ionization source in negative-ion mode (Agilent 6460 Triple Quadrupole System). The isotope dilution method was used for quantification using a five-point calibration curve (0.5, 2.5, 10, 40, 80 and 400 pg absolute injected). The average recoveries in the cell lysates were in the range of 78–98% (Supporting information, Table S2). Details on the analytical method are described previously (Ahrens, Norström, Viktor, Cousins, & Josefsson, 2015).

2.6 | Protein determination

Protein content was determined by the bicinchoninic acid (BCA) assay in lysates from cells for PPARα activity determination as well as cells for cellular PFAS quantification to enable comparison between activity and concentration. A protein standard curve was prepared (Sigma-Aldrich) in 25% passive lysis buffer (Promega). The standards and cell lysates (2.5 μl per well in duplicates) were added to 96-well plates and diluted 1:4 with MilliQ water. BCA protein assay reagent, 190 μl (1:50, copper sulfate:BCA solution; Sigma-Aldrich), was added and incubated at 37°C for 30 minutes. Absorbance was measured at 570 nm (Wallac Victor 1420 microplate reader; PerkinElmer).

2.7 | Data processing and statistical analysis

The relative light units for PPARα activity was determined by normalization of the firefly to the renilla luminescence in the well. Successively, PPARα activity was normalized to the means of the controls after adjustment for protein levels in the respective well. The PFAS cellular concentrations were normalized to the protein level in the pooled wells.

The PPARα activity data were analyzed by Kruskal–Wallis to detect differences among treatment groups followed by a Dunn's post-hoc test to assess differences between controls and the various treatment groups. A non-parametric test was selected as some treatment groups had a small number of replicates ($n = 3$) and thus normality was not assessed. A two-sided Jonckheere's trend test was used to examine possible concentration–response relationships between PPARα activity and cellular concentration. All PFASs exhibiting a statistically significant positive relationship in the Jonckheere's trend test were fitted by linear regression. Finally, Mann–Whitney was performed for pair-wise comparison between PFCAs and PFSAs with the same perfluorocarbon chain lengths, but different functional groups, to assess statistically significant differences in PPARα activities. The level of significance was set at $P < 0.05$. All graphs and statistical analyses were conducted in GraphPad Prism 5.02, except the Jonckheere's trend test, which was performed in StatXact 10.0.

3 | RESULTS

3.1 | Cell viability

Of the tested PFASs, only FOSA led to more than 20% decrease in cell viability at 100 μM and thus this concentration was not included in further testing. A few other statistically significant effects on cell viability were observed, although sporadic and not concentration dependent (Supporting information, Figure S1).

3.2 | Protein content

The total protein content was generally stable across different PFAS treatments in cells used for PPARα activity and cellular concentration determination (Supporting information, Figure S2). A slight but statistically significant decrease in protein content was observed at 100 μM of PFDA and PFTeDA.

3.3 | Cellular perfluoroalkyl substance concentrations

All tested PFASs were detected in cell lysates in concentrations above the limit of quantification in all treatment groups. The fraction of PFASs taken up by the cells was calculated as the total moles detected

in the cells divided by the total moles added to the well, and expressed as percentages (Table 1). The uptake percentage was not dependent on the medium concentrations of PFASs and was below 1%, except for PFDoDA, PFTeDA and FOSA, for which uptake between 1.21 and 4.06% was observed (Table 1).

The cellular PFAS concentrations, expressed as nmol mg^{-1} protein, for the 14 PFASs at treatment concentrations of 10, 30 and 100 μM are shown in Figure 2(A). The concentrations ranged 18–2490 nmol mg^{-1} protein for the PFCAs, 4–494 nmol mg^{-1} protein for the PFSAs, and were 81 and 435 nmol mg^{-1} protein for FOSA. The cellular concentrations increased with increasing treatment concentrations of PFASs. The highest cellular concentrations of the PFCAs were observed after treatment of cells with the long-chain PFUnDA, PFDoDA and PFTeDA, whereas the lowest cellular concentrations were seen after treatment with the short-chain PFBA (Figure 2A). This pattern was not observed for the PFSAs, for which PFHxS had the highest cellular concentration both at 30 and 100 μM, followed by PFBS, and PFOS with the lowest cellular concentration.

To study the impact of the functional group on cellular concentrations, we compared PFASs with the same perfluorocarbon chain lengths, but with different functional groups. Though the conventional PFAS nomenclature is used in this paper, which is based on the total number of carbon atoms in the molecule, a comparison based on perfluorocarbon chain length is imperative to obtain information on the structure–activity relationships. Thus, PFPeA, PFHpA and PFNA were compared to PFBS, PFHxS and PFOS, respectively. The cellular concentration was higher for the PFCAs than the PFSAs, except for PFPeA compared to PFBS at 10 μM and PFHpA compared to PFHxS at 100 μM. FOSA had higher cellular concentration compared to the two compounds with the same perfluorocarbon chain length, but different functional groups: carboxylate (PFNA) and sulfonate (PFOS) (Figure 2A).

TABLE 1 PFAS uptake in HepG2 cells expressed as percentage total moles detected in the cells of total moles added to the wells at treatment concentrations of 10, 30 and 100 μM for 24 hours. Values are based on three pooled wells for each treatment concentration.

	Cellular uptake (%)		
	10 μM	30 μM	100 μM
PFBA	0.14	0.13	0.10
PFPeA	0.18	0.26	0.23
PFHxA	0.19	0.19	0.14
PFHpA	0.16	0.22	0.21
PFOA	0.24	0.27	0.33
PFNA	0.15	0.22	0.31
PFDA	0.23	0.23	0.45
PFUnDA	0.61	0.54	0.84
PFDoDA	1.21	1.33	1.65
PFTeDA	4.06	1.73	0.55
PFBS	0.25	0.19	0.15
PFHxS	0.08	0.18	0.33
PFOS	0.04	0.04	0.05
FOSA	0.91	1.35	–

(–) not tested due to compromised cell viability. See text for definitions of abbreviations.

3.4 | Peroxisome proliferator-activated receptor alpha activity

All PFASs induced PPARα activity statistically significantly, as compared to the vehicle control, except PFUnDA, PFOS and FOSA (Figure 2B). The lowest observed effect concentrations (LOECs), expressed as nominal concentrations, causing statistically significant PPARα activation were 30 or 100 μM, except PFTeDA for which an effect was observed from 10 μM. For all treatment concentrations, there was an apparent increase in PPARα activity with PFCA perfluorocarbon chain length, which peaked with PFOA. Short-chain (PFBA and PFPeA) and long-chain (PFDoDA and PFTeDA) PFCAs induced PPARα activity up to twofold. Similar fold-induction was observed for PFBS and PFHxS. The highest induction of PPARα activity, 2.5–3.7-fold, was observed after treatment of cells with PFHxA, PFHpA, PFOA, PFNA and PFDA.

To study the impact of the functional group on PPARα activity, we compared PFASs with the same perfluorocarbon chain lengths, but with different functional groups. Thus, PFPeA, PFHpA and PFNA were compared to PFBS, PFHxS and PFOS, respectively (Figure 2B). The PPARα activity was generally higher for PFCAs than PFSAs with the same perfluorocarbon chain length, a difference that was statistically significant for PFNA compared to PFOS at 30 and 100 μM ($P < 0.05$).

3.5 | Peroxisome proliferator-activated receptor alpha activity and cellular concentrations

Of the 11 PFASs exhibiting statistically significant activation of PPARα, seven had LOECs in the range 38–164 nmol mg^{-1} protein, expressed as cellular concentrations (Figure 2A,B). These PFASs included PFCAs with chain lengths up to and including PFOA, as well as PFBS and PFHxS. The remaining four PFASs had LOECs ranging from 458 to 2490 nmol mg^{-1} protein and this group contained the long-chain PFCAs, PFNA, PFDA, PFDoDA and PFTeDA. The maximum induction of PPARα activity was similar for short-chain (PFBA and PFPeA) and long-chain PFCAs (PFDoDA and PFTeDA) (approximately twofold), but PPARα activities were induced at lower cellular concentrations for the short-chain homologs compared to the long-chain homologs (Figure 2A,B).

A statistically significant positive relationship between PPARα activity and cellular PFAS concentration was demonstrated for all PFASs, except PFOS and FOSA. For FOSA there was a statistically significant negative relationship. Linear regression was performed for all PFASs with a statistically significant positive relationship (Figure 3). In general, the linear relationships exhibited a continuous decrease in steepness with increasing chain length for PFCAs, indicating a higher potency of short-chain PFCAs (Figure 3A). PFBS and PFHxS displayed flat relationships (Figure 3B). Estimated slopes and r^2 values obtained from linear regression are given in Table 2. The r^2 values were ≥0.5, except for PFUnDA, PFBS and PFHxS. The estimated slopes were significantly different from zero and were higher for PFBA, PFPeA, PFHxA, PFHpA, PFOA and PFNA, than for PFDA, PFUnDA, PFDoDA, PFBS and PFHxS. Slopes were decreasing from PFHxA to PFDoDA, supporting the general observation in Figure 3 of a continuous decrease in steepness with increasing chain length.

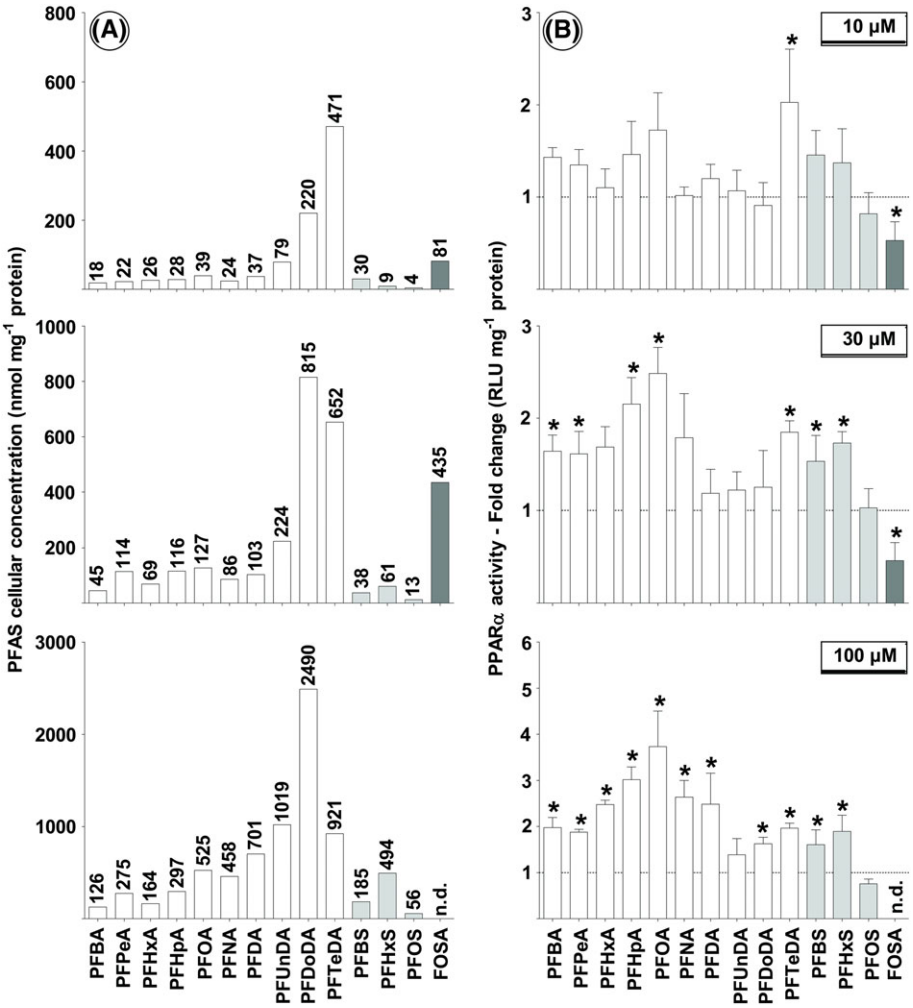


FIGURE 2 Cellular PFAS concentrations (nmol mg^{-1} protein) (a) and PPARα activity (fold-change compared to controls (dotted lines), RLU mg^{-1} protein) (B) in HepG2 cells treated with 10, 30 and 100 µM of PFASs for 24 hours. Results are shown in the order of increasing perfluorocarbon chain length. White = PFCAs, light gray = PFSA; dark gray = FOSA. Bars showing cellular concentrations are based on three pooled wells for each treatment concentration (a) and bars showing PPARα activity are means ± SD from three to four replicates for PFAS treatment and seven to eight replicates for vehicle controls (B). N.D., not determined as FOSA was cytotoxic at 100 µM. *significant difference compared to control ($P < 0.05$). See text for definitions of abbreviations

4 | DISCUSSION

The present study in HepG2 cells demonstrates for the first time relationships between PPARα activity and cellular concentration of a broad range of PFASs with varying perfluorocarbon chain lengths and functional groups. The relationships were positive for all the tested compounds, except PFOS and FOSA, and were compound-specific, as demonstrated by differences in the estimated slopes. The concentration–response relationships were steeper for PFCAs with chain lengths up to and including PFNA than for the other studied PFASs. This indicates that the PPARα activity is not only dependent on the cellular concentration, but also varies with the chemical characteristics of the PFASs.

The percentage uptake of PFASs from the medium was low and varied from 0.04 to 4.1%. This may seem low; however, the absolute cellular concentrations of 4–2500 nmol mg^{-1} protein was sufficient to induce PPARα activity and somewhat higher than reported previously in other cell lines (Gorrochategui, Pérez-Albaladejo, Casas, Lacorte, & Porte, 2014; Nakagawa et al., 2007; Weaver, Ehresman, Butenhoff, & Hagenbuch, 2010). Our results demonstrate that cellular concentrations are affected by the functional group and the perfluorocarbon chain length. For the PFCAs the long-chain PFUnDA, PFDoDA and PFTeDA had the highest cellular concentration and the short-chain PFBA had the lowest cellular concentration in all treatment groups. To our knowledge, there are no studies on cellular uptake of PFASs in liver cell lines. However, in JEG-3 cells treated with a mixture of eight PFASs, similar effects of chain length on cellular uptake have been demonstrated (Gorrochategui et al., 2014). The order of PFASs ranked by cellular concentrations were PFDoDA > PFOS> > PFNA > PFOA > PFHxA, while the short-chain PFBA, PFBS and PFHxS were not detected (Gorrochategui et al., 2014). Similarly, Kudo, Suzuki-Nakajima, Mitsumoto, and Kawashima (2006) reported increasing liver concentrations with increasing PFCA chain length in mice intraperitoneally administered with PFHxA, PFHpA, PFOA or PFNA for 5 days, which was positively correlated with liver peroxisomal β-oxidation (Kudo et al., 2006).

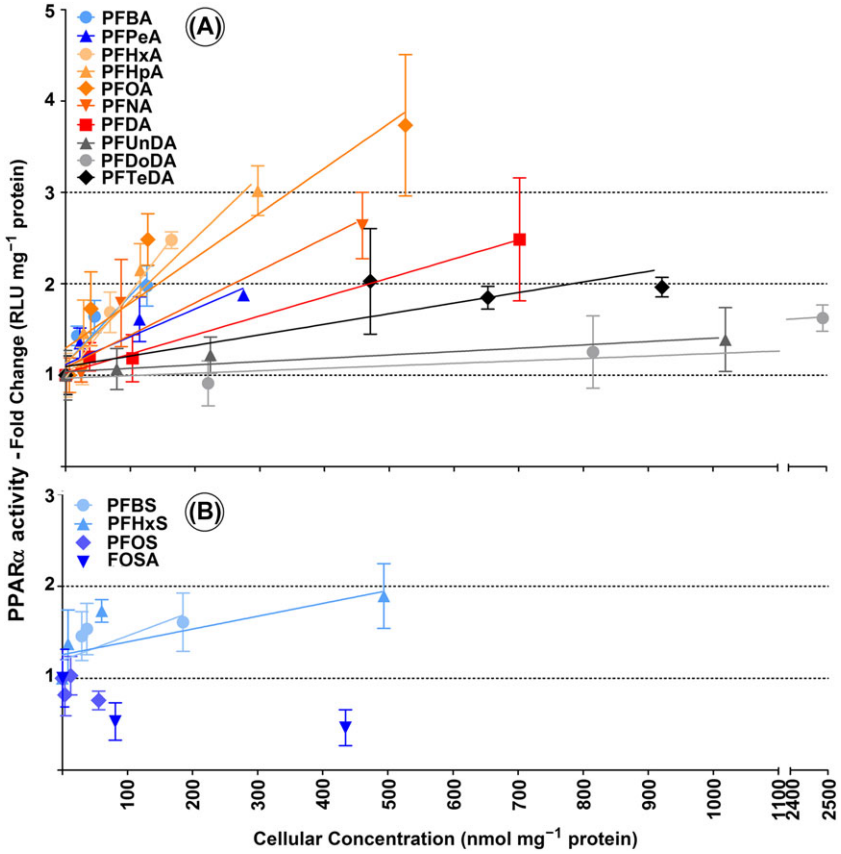


FIGURE 3 Relationship between PPARα activity (fold-change compared to control (RLU mg^{-1} protein) and cellular concentrations (nmol mg^{-1} protein) of PFCAs (A) and PFSA and FOSA (B). A statistically significant positive relationship was demonstrated between PPARα activity and cellular concentrations for all PFASs, except PFOS and FOSA by Jonckheere's trend test. Linear regression was performed on all statistically significant positive relationships. Based on data from Figure 2. See text for definitions of abbreviations

TABLE 2 Estimated slopes and r^2 values based on linear regression on protein adjusted PPARα activity and cellular concentration

	Slope	r^2
PFBA	7.4×10^{-3}	0.69
PFPeA	3.0×10^{-3}	0.67
PFHxA	9.3×10^{-3}	0.89
PFHpA	6.8×10^{-3}	0.86
PFOA	4.9×10^{-3}	0.78
PFNA	3.5×10^{-3}	0.78
PFDA	2.1×10^{-3}	0.74
PFUnDA	0.4×10^{-3}	0.25
PFDoDA	0.3×10^{-3}	0.49
PFTeDA	1.2×10^{-3}	0.61
PFBS	2.6×10^{-3}	0.28
PFHxS	1.4×10^{-3}	0.39
PFOS	–	–
FOSA	–	–

–, no positive statistically significant relationship by the Jonckheere trend test.

All slopes were statistically significant different from zero. See text for definitions of abbreviations.

Physico-chemical characteristics and molecular processes may explain differences in cellular uptake of PFASs. As the tested PFASs are not metabolized, the variations in cellular concentrations are not due to differences in metabolism. Increasing perfluorocarbon chain length is associated with increasing partitioning coefficients (Hidalgo & Mora-Diez, 2016; Wang, MacLeod, Cousins, Scheringer, & Hungerbühler, 2011), which may explain the high cellular concentrations of long-chain PFCAs demonstrated in our study. Furthermore, PFASs can traverse the cell membrane either by simple diffusion, as shown for PFOA in isolated rat hepatocytes (Han, Yang, Snajdr, Nabb, & Mingoia, 2008) or uptake by specific transporter proteins, as shown for some PFASs (Nakagawa et al., 2007; Weaver et al., 2010; Yang, Glover, & Han, 2010; Zhao et al., 2017). Another factor, which may affect the cellular uptake, is interaction with serum proteins (Bischel, MacManus-Spencer, & Luthy, 2010; Bischel, Macmanus-Spencer, Zhang, & Luthy, 2011; D'eon, Simpson, Kumar, Baer, & Mabury, 2010).

PPARα activity was increased statistically significantly, except after treatment with PFUnDA, PFOS and FOSA. The magnitude of PFCA-induced activity increased with chain length up to and including PFOA and the PFCAs generally exhibited greater activities than the PFSAs. Our results on human PPARα activities in transfected HepG2 cells are generally in line with previous reports on effects of PFSAs and PFCAs in transfected HEK293 and COS-1 cells. Thus, most of the examined compounds have previously been reported to activate human PPARα, an increase in potency has been reported up to and including PFOA or PFNA, and PFCAs are generally more potent than the PFSAs (Buhrke et al., 2013; Takacs & Abbott, 2007; Wolf et al., 2012, 2008). Wallace, Kissling, Melnick, and Blystone (2013), who studied effects on rat liver mitochondrial respiration of 16 PFASs, also demonstrated, similar to our results, an increase in potency with increasing chain lengths of PFASs with a prominent inflection point between the six- and eight-carbon congeners (Wallace et al., 2013). To the best of our knowledge, there are no previous reports on PPARα activity of FOSA, which in the present study did not increase the

PPARα activity, and of PFTeDA, which induced PPARα activity even at the lowest tested concentration of 10 μM.

Interestingly, despite the higher cellular concentrations of long-chain PFCAs, they caused lower activity than PFHxA, PFHpA, PFOA and PFNA. This may reflect differences in receptor affinity and efficacy leading to differences in activity, but may also be the result of long-chain PFCA cell membrane binding, rather than an actual uptake into the cell. PFASs are amphiphilic and can incorporate into cell membranes, which are composed of amphiphilic phospholipids. In line with this, Kasuya, Nakano, Katayama, and Hatanaka (2011) reported that liposome membrane affinity increased with increasing chain length of perfluoroalkyl glycosides and suggested that the glycosides with high fluorine content remained embedded in the cell membrane (Kasuya et al., 2011). Additionally, long-chain PFCAs have been reported to be transported to the nucleus to a lower extent by human liver fatty acid binding protein compared to short-chain PFCAs, which may further explain the lower PPARα activity (Zhang, Ren, & Guo, 2013).

Overall we found that: (1) all the tested PFASs were detected in the cells with a low percentage cellular uptake; (2) the cellular concentration increased with increasing chain lengths for the PFCAs; (3) when comparing PFASs with the same perfluorocarbon chain length, the cellular concentration was highest for FOSA, followed by PFCAs and lastly PFSAs; (4) all PFCAs, except PFUnDA, induced PPARα activity with potencies increasing up to and including PFOA; (5) the PPARα activity was higher for PFCAs than PFSAs; (6) there was a significant positive relationship between the PPARα activity and the cellular PFAS concentration except for PFOS and FOSA; and (7) the relationships were compound specific.

To our knowledge, this is the first report of cellular concentration in HepG2 cells of a broad range of PFASs of varying perfluorocarbon chain lengths and functional groups and the first report establishing a relationship between PPARα activity and cellular concentration. Most in vitro studies do not include measurements of cellular concentrations, and conclusions on dose–response relationships are usually based on the added nominal concentration of the test substance. By measuring cellular concentrations, it was revealed that short-chain PFASs exhibited activity at low cellular concentrations, which is of particular interest as they have replaced long-chain PFASs for some applications. Thus, the present investigation demonstrates that combining chemical analysis of cellular concentration with activity measurements allows a more exhaustive analysis of concentration–response relationships from in vitro studies.

CONFLICT OF INTEREST

The authors did not report any conflict of interest.

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ORCID

Anna Kjerstine Rosenmai http://orcid.org/0000-0002-0588-607X
Lutz Ahrens http://orcid.org/0000-0002-5430-6764
Johan Lundqvist http://orcid.org/0000-0001-5693-9007
Agneta Oskarsson http://orcid.org/0000-0002-3134-7811

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